HARD ALGEBRA

Algebra

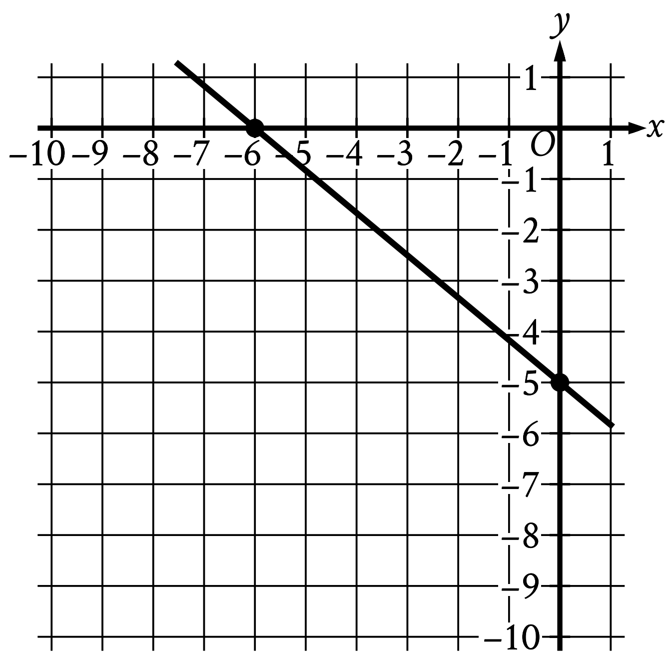
10 questions · ~15 min

01

SAT QUESTION BANK 2026-05-29

Q1

GRID-IN ALGEBRA



- The line slants gradually down from left to right.
- The line passes through the following points:
  - (negative 6 comma 0)
  - (0 comma negative 5)

Line k is shown in the xy-plane. Line j (not shown) is perpendicular to line k. What is the slope of line j?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5

6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q2 GRID-IN ALGEBRA

For the function f, $f c x = x - 8$ for all values of x, where c is a positive constant. If $f 2 = 35$, what is the value of c?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3 ALGEBRA

A team hosting an event to raise money for new uniforms plans to sell at least 140 tickets before this event and at least 220 tickets during this event to raise a total of at least $\$ 5{,}820$ from all tickets sold. The price of a ticket during this event is $\$ 3$ less than the price of a ticket before this event. Which inequality represents this situation, where x is the price, in dollars, of a ticket sold during this event?

A. $140x + 3 + 220x \leq 5{,}820$

B. $140x + 3 + 220x \geq 5{,}820$

C. $140x - 3 + 220x \leq 5{,}820$

D. $140x - 3 + 220x \geq 5{,}820$

Q4

ALGEBRA

If $\frac{x - 5}{7} = \frac{x - 5}{9}$, the value of x - 5 is between which of the following pairs of values?

A. -9 and -7

B. -3 and 3

C. 4.5 and 5.5

D. 6.75 and 9.25

Q5

GRID-IN ALGEBRA

$$\frac{7}{8}y - \frac{5}{8}x = \frac{4}{7} - \frac{7}{8}y$$

$$\frac{5}{4}x + \frac{7}{4} = py + \frac{15}{4}$$

In the given system of equations, p is a constant. If the system has no solution, what is the value of p?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

ALGEBRA

Line ℓ in the xy-plane is perpendicular to the line with equation $x = 2$. What is the slope of line ℓ?

A. 0

B. $-\frac{1}{2}$

C. -2

D. The slope of line ℓ is undefined.

Q7 ALGEBRA

x	fx
-4	0
$-\frac{19}{5}$	1
$-\frac{18}{5}$	2

For the linear function f, the table shows three values of x and their corresponding values of fx. If $hx = fx - 13$, which equation defines h?

A. $hx = 5x - 4$

B. $hx = 5x + 7$

C. $hx = 5x + 9$

D. $hx = 5x + 20$

Q8

GRID-IN ALGEBRA

A local transit company sells a monthly pass for $95 that allows an unlimited number of trips of any length. Tickets for individual trips cost $1.50, $2.50, or $3.50, depending on the length of the trip. What is the minimum number of trips per month for which a monthly pass could cost less than purchasing individual tickets for trips?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

ALGEBRA

How many solutions does the equation $12x - 3 = -3x + 12$ have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

Q10

GRID-IN ALGEBRA

$$4x - 6y = 10y + 2$$

$$ty = \frac{1}{2} + 2x$$

In the given system of equations, t is a constant. If the system has no solution, what is the value of t?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
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Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.